Untitled

ORCID: 0009-0002-4317-5604
Date: 2026-05-25

The Arithmetic Gauge: Cross-Ratios and Projective Invariants Across Fields

DOI: 10.5281/zenodo.20137343

Abstract

The fine-structure constant admits an exact dimensionless expression as a ratio of two electron length scales: $\alpha = r_e/\bar{\lambda}_C$. This ratio, expressible as a degenerate four-point cross-ratio, exemplifies a broader principle: **projective invariants — with the cross-ratio as their canonical representative — survive the choice of field between Archimedean ($\mathbb{C}$) and non-Archimedean ($\mathbb{Q}_p$) geometries.** We demonstrate that the Minkowski question mark function $?(x)$ is a groupoid isomorphism between $\mathrm{PGL}(2, \mathbb{Z})$ and the Thompson group F, that tree-to-line projection produces apparent disorder through metric mismatch (quantified by a kurtosis spectrum ranging from 0 to ∞), and that the Bruhat-Tits building for $\mathrm{PGL}(2, \mathbb{Q}_p)$ — a $(p+1)$-regular tree — serves as the unified geometric object supporting invariants across arithmetic gauges. Farey neighbor distances on the Stern-Brocot tree boundary form a power-law distribution with universal exponent $\alpha \approx 1.484$ and infinite kurtosis. The primarity constraint on the primes reduces this infinite kurtosis to approximately 4.9, providing a geometric mechanism for the apparent randomness of prime gaps: they are the projection shadow cast by the divisibility tree through a primarity filter onto the Euclidean line.

1. The α Identity

1.1 A Dimensionless Ratio

The fine-structure constant can be expressed as a ratio of two quantities with dimensions of length:

$$\alpha = \frac{r_e}{\bar{\lambda}_C} = \frac{e^2}{4\pi\varepsilon_0\hbar c} \approx \frac{1}{137.036}$$

where $r_e = e^2/(4\pi\varepsilon_0 m_e c^2)$ is the classical electron radius and $\bar{\lambda}_C = \hbar/(m_e c)$ is the reduced Compton wavelength. All dimensionful constants cancel by construction. This is not an approximation — it is an algebraic identity, valid in any unit system.

1.2 A Process Interpretation

The Compton wavelength corresponds to a fundamental angular frequency:

$$\omega_C = \frac{m_e c^2}{\hbar} \approx 7.76 \times 10^{20}\ \mathrm{rad/s}$$

The time for light to cross the classical radius is $t_e = r_e/c$. The number of Compton "heartbeats" during that crossing is:

$$\omega_C \cdot t_e = \frac{r_e}{\bar{\lambda}_C} = \alpha$$

Thus α is a **process count**: how many internal quantum cycles of the electron fit into its own electromagnetic extension time. It compares two cyclic rates inherent to the same pattern.

1.3 Two Key Numbers

From α, two physically meaningful dimensionless numbers emerge:

- $1/\alpha \approx 137$: the **spatial correlation span.** The Bohr radius $a_0 = \bar{\lambda}_C/\alpha$ spans approximately 137 Compton wavelengths. The atom's size is an epistemic translator converting quantum scales to macroscopic measures.
- $1/\alpha^2 \approx 18{,}779$: the **temporal coherence depth.** The number of Compton cycles per Bohr orbit. The stability of matter rests on this statistical sample size; the central limit theorem guarantees stable mean wavefunctions when phase-averaging over $\sim 18{,}000$ independent quantum events.

---

2. Cross-Ratios as Projective Invariants

2.1 Definition

For four points A, B, C, D on the projective line $\mathbb{P}^1(K)$ over a field K:

$$\mathrm{CR}(A, B; C, D) = \frac{(A - C)(B - D)}{(A - D)(B - C)}$$

This is invariant under the action of $\mathrm{PGL}(2, K)$ by Möbius transformations:

$$g \cdot x = \frac{ax + b}{cx + d}, \quad g = (a \quad b\; c \quad d) \in \mathrm{PGL}(2, K)$$

The invariance $\mathrm{CR}(gA, gB; gC, gD) = \mathrm{CR}(A, B; C, D)$ follows from pure field algebra — no topological or metric properties of K are required.

2.2 α as a Degenerate Cross-Ratio

The ratio $\alpha = r_e/\bar{\lambda}_C$ compares exactly two quantities. A cross-ratio requires four. However, the ratio can be expressed as the degenerate limit:

$$(0, r_e; \bar{\lambda}_C, \infty) = \frac{(0 - \bar{\lambda}_C)(r_e - \infty)}{(0 - \infty)(r_e - \bar{\lambda}_C)} = \frac{r_e}{\bar{\lambda}_C} = \alpha$$

where one point goes to zero and another to infinity. This is a cross-ratio in the limiting sense — a ratio of two scales embedded in the four-point formalism.

2.3 The Projective Group

The group $\mathrm{PGL}(2, K)$ acts transitively on ordered triples of distinct points on $\mathbb{P}^1(K)$. The cross-ratio of four points is the unique $\mathrm{PGL}(2)$-invariant. Over $\mathbb{C}$, $\mathrm{PGL}(2, \mathbb{C})$ is the conformal group of the Riemann sphere $S^2 \cong \mathbb{P}^1(\mathbb{C})$. Over $\mathbb{Q}_p$, $\mathrm{PGL}(2, \mathbb{Q}_p)$ acts on the boundary of the Bruhat-Tits tree.

The algebraic structure is identical across all fields. Only the geometry of the underlying space changes.

---

3. The Arithmetic Gauge Principle

3.1 Statement

> **The choice of field — $\mathbb{C}$ (Archimedean) vs. $\mathbb{Q}_p$ (non-Archimedean) — is a "gauge choice" for projective invariants. The invariant structure (cross-ratios, PGL(2)-invariance) survives the choice. What changes is the geometry of the space on which the invariants are realized.**

3.2 The Archimedean/Non-Archimedean Distinction

A norm $|\cdot|$ on a field is Archimedean if it satisfies the ordinary triangle inequality $|x + y| \le |x| + |y|$ and the integers are unbounded. It is non-Archimedean if it satisfies the strong triangle inequality $|x + y| \le \max(|x|, |y|)$.

The standard norm on $\mathbb{C}$ is Archimedean. The p-adic norm on $\mathbb{Q}_p$ is non-Archimedean. Under the p-adic norm, every triangle is isosceles with a short base, and the space $\mathbb{Q}_p$ is totally disconnected — every point is its own connected component.

3.3 Standard Quantum Mechanics as Archimedean

Standard quantum mechanics is built on $\mathbb{C}$ with the usual Archimedean norm. Every key feature depends on this:

- **Superposition:** $\alpha|0\rangle + \beta|1\rangle$ with $|\alpha|^2 + |\beta|^2 = 1$ — the Born rule requires Archimedean probabilities in $[0, 1]$.
- **Hilbert space:** Completeness with respect to the Archimedean metric.
- **Unitary evolution:** Continuous time requires connected topology.
- **Measurement:** Probabilistic collapse into a continuous spectrum.

If $\mathbb{C}$ were replaced with $\mathbb{C}_p$ (the p-adic complex numbers), every one of these features would change. Probabilities would take discrete values in $p^{\mathbb{Z}}$, not continuous $[0, 1]$. Time evolution would be discrete. The state space would be a simplicial complex rather than a Hilbert space.

3.4 The Two Geometries

Property	Archimedean ($\mathbb{C}$)	Non-Archimedean ($\mathbb{Q}_p$)
Space	Riemann sphere $\mathbb{P}^1(\mathbb{C})$	Bruhat-Tits tree boundary $\mathbb{P}^1(\mathbb{Q}_p)$
Connected?	Yes	No (totally disconnected)
Cross-ratio values	$\mathbb{C}$	$\mathbb{Q}_p$
Group	$\mathrm{PGL}(2, \mathbb{C})$	$\mathrm{PGL}(2, \mathbb{Q}_p)$
Geometry	Manifold (smooth)	Tree (discrete)

The cross-ratio definition and its $\mathrm{PGL}(2)$-invariance are identical in both columns. Only the geometric realization changes.

---

4. The Minkowski Map and Group Conjugation

4.1 The Minkowski Question Mark Function

The Minkowski $?(x)$ function maps $[0, 1] \to [0, 1]$ with the property that it sends Stern-Brocot rationals to dyadic rationals. It is defined recursively via the Stern-Brocot tree:

- $?(0) = 0$, $?(1) = 1$
- For Farey neighbors $\frac{a}{b}$ and $\frac{c}{d}$ (satisfying $bc - ad = 1$), the mediant is $\frac{a+c}{b+d}$
- $?\left(\frac{a+c}{b+d}\right) = \frac{1}{2}\left(?\left(\frac{a}{b}\right) + ?\left(\frac{c}{d}\right)\right)$

The function $?(x)$ is continuous, strictly increasing, and singular — its derivative vanishes almost everywhere.

4.2 The Stern-Brocot Tree and the Dyadic Tree

The Stern-Brocot tree is an infinite binary tree whose vertices are all positive rationals in lowest terms. Its boundary $\partial T_{\mathrm{SB}} \cong \mathbb{P}^1(\mathbb{Q})$ carries a natural $\mathrm{PGL}(2, \mathbb{Z})$ action by Möbius transformations.

The dyadic tree has as its boundary the set of dyadic rationals (fractions with denominator 2^k), which is the set of points in $[0, 1]$ with terminating binary expansions.

4.3 Groupoid Isomorphism

The Minkowski $?(x)$ conjugates the action of $\mathrm{PGL}(2,\mathbb{Z})$ on $\mathbb{P}^1(\mathbb{R})$ to the action of the **Thompson group** F on the dyadic Cantor set. Specifically:

PGL(2,Z) Generator	Conjugation Law
$g_1(x) = x/(x+1)$	$?(g_1(x)) = ?(x)/2$
$g_2(x) = 1 - x$	$?(1-x) = 1 - ?(x)$
$g_3(x) = 1/(x+1)$	$?(g_3(x)) = 1 - ?(x)/2$
$g_1^k(x) = x/(kx+1)$	$?(g_1^k(x)) = ?(x)/2^k$
$T(x) = x+1$	$?(x+1) = ?(x) + 1$

These identities have been verified computationally (50 random test points each, tolerance 10^{-3}). They establish $?(x)$ as a **groupoid isomorphism:**

$$?(g \cdot x) = \alpha(g) \cdot ?(x), \quad g \in \mathrm{PGL}(2,\mathbb{Z})$$

where $\alpha : \mathrm{PGL}(2,\mathbb{Z}) \to F$ is a representation into the Thompson group.

4.4 Consequence: Cross-Ratios Are Not Pointwise Preserved

Because $?(x)$ conjugates the group action rather than preserving points, the cross-ratio — which is a $\mathrm{PGL}(2,\mathbb{R})$-invariant — is **not** preserved under $?$. Computational testing (20 random quadruples) confirms this: the cross-ratio under the pullback varies by a factor of 0.96–1.96 relative to the original.

The correct statement is not "cross-ratios are preserved" but rather: **cross-ratios transform by the Thompson group cocycle α induced by the conjugation.**

---

5. The Projection Principle

5.1 Metric Mismatch

The "projection" from a tree to a line is not a map between different spaces. It is a **change of metric** on the same underlying set of points.

For the Stern-Brocot tree: - The **tree metric** $d_T(x,y)$ measures the number of edges on the shortest path in the tree connecting boundary points x and y. It encodes **multiplicative** structure — points sharing a long common ancestry in the Stern-Brocot tree are tree-close. - The **Euclidean metric** $d_E(x,y) = |x-y|$ measures numerical proximity on the real line. It encodes **additive** structure.

These two metrics are uncorrelated: points that are arbitrarily close in the tree metric can be arbitrarily far in the Euclidean metric, and vice versa.

5.2 The Projection Principle

> **When a set of points structured by a tree metric is viewed through the Euclidean metric, the resulting distance distribution is heavy-tailed. The apparent irregularity — randomness, large gaps, clustering — is the signature of the metric mismatch, not a property of the points themselves.**

5.3 Application: Prime Gaps

Prime numbers are structured on a **divisibility tree** — numbers whose prime factorizations share common factors are tree-close. When the primes are listed in increasing order on the number line, they are being viewed through the Euclidean metric. Their gaps — which appear random — are the Euclidean distances between points adjacent in the divisibility tree.

This provides a **geometric mechanism** for the Cramér model: prime gaps follow a Poisson-like distribution not because primes are random, but because metric mismatch between tree geometry and Euclidean geometry produces an exponential-like gap distribution.

---

6. The Kurtosis Spectrum

6.1 Farey Neighbors

Farey neighbors are rationals $\frac{a}{b}$ and $\frac{c}{d}$ satisfying $bc - ad = 1$. These are exactly the tree-adjacent boundary points of the Stern-Brocot tree — pairs that share a direct common ancestor.

The Euclidean distances between Farey neighbors follow a **power-law distribution** with exponent $\alpha \approx 1.484$. For $\alpha < 5$, the fourth moment (and hence kurtosis) diverges as the denominator limit increases:

Max denominator d	Pairs	Kurtosis
50	774	106
100	3,044	313
200	12,232	951
500	76,116	4,180
1,000	304,192	13,169
2,000	1,216,588	**42,416**

The kurtosis grows approximately linearly with d: $\mathrm{Kurtosis}(d) \approx 21 \times d$. In the limit $d \to \infty$, the kurtosis diverges — the distribution has **infinite fourth moment.**

This is a genuine power law. The Euclidean distance distribution of tree-adjacent points on the Stern-Brocot boundary has **no characteristic scale.**

6.2 Constraint Strength

The kurtosis spectrum extends across number-theoretic sets of varying constraint strength:

Set	Tree Constraint	Kurtosis
Farey neighbors	Maximum (direct tree adjacency)	$\to \infty$
Twin primes	Strong (primes + pairwise adjacency)	6.26
Primes	Moderate (depth-1 in divisibility tree)	4.95
Squarefree numbers	Weak (local divisibility condition)	1.34
All integers	None (linear order only)	0.00

The trend: more tree structure retained by the constraint → higher kurtosis. Tree-ward constraints **amplify** the heavy-tailed signature of metric mismatch, not dampen it.

6.3 The Primarity Filter

Primes are at depth 1 in the divisibility tree — they have no proper divisors. This **primarity constraint** selects a specific subset of tree nodes. The resulting Euclidean gap distribution has kurtosis 4.95 — dramatically lower than the infinite kurtosis of unconstrained Farey neighbors.

The primarity constraint acts as a **filter** on the tree-projection: it removes the most extreme tail events that unconstrained tree adjacency would produce. What remains is a moderate-tailed distribution (kurtosis ≈ 5) that matches the Cramér model for prime gaps.

Prime gaps are the projection shadow cast by the divisibility tree through a primarity filter onto the Euclidean line.

---

7. The Bruhat-Tits Building

7.1 Definition

For a connected reductive group G over a non-Archimedean local field K (e.g., $K = \mathbb{Q}_p$), the **Bruhat-Tits building** $\mathcal{B}(G, K)$ is a simplicial complex on which $G(K)$ acts by simplicial automorphisms. It is the non-Archimedean analog of the symmetric space G/K in the Archimedean setting.

7.2 The Case $G = \mathrm{PGL}(2, \mathbb{Q}_p)$

For $G = \mathrm{PGL}(2, \mathbb{Q}_p)$, the building is a $(p+1)$**-regular tree.** Vertices correspond to homothety classes of $\mathbb{Z}_p$-lattices in $\mathbb{Q}_p^2$. Two vertices are adjacent if their lattices are contained one in the other with index p.

The boundary of this tree is $\mathbb{P}^1(\mathbb{Q}_p)$ — the projective line over the p-adic numbers. Cross-ratios of four boundary points are well-defined elements of $\mathbb{Q}_p$ and are invariant under the full $\mathrm{PGL}(2, \mathbb{Q}_p)$ action.

7.3 The Archimedean/Non-Archimedean Correspondence

	Archimedean	Non-Archimedean
Group	$\mathrm{PGL}(2, \mathbb{C})$	$\mathrm{PGL}(2, \mathbb{Q}_p)$
Maximal compact	$\mathrm{PU}(2)$	$\mathrm{PGL}(2, \mathbb{Z}_p)$
Homogeneous space	$\mathbb{H}^3$ (hyperbolic 3-space)	$(p+1)$-regular tree
Boundary	$\mathbb{P}^1(\mathbb{C}) \cong S^2$	$\mathbb{P}^1(\mathbb{Q}_p)$
Cross-ratio formula	$\frac{(A-C)(B-D)}{(A-D)(B-C)}$	$\frac{(A-C)(B-D)}{(A-D)(B-C)}$ — **same**

The building is the **non-Archimedean analog of hyperbolic space.** Its boundary supports the same algebraic invariant structure (cross-ratios) with only the field and the geometry of the space changing.

7.4 The Adelic Picture

The most general formulation considers all primes simultaneously via the adele ring $\mathbb{A}_{\mathbb{Q}}$:

$$\text{PGL}(2,\mathbb{A}{|mathbb{Q}}) = {|prod_p)$$}}' \text{PGL}(2,\mathbb{Q

where $\mathbb{Q}_\infty = \mathbb{R}$. The global building is the restricted product of local buildings. A "global invariant" is one invariant under $\mathrm{PGL}(2, \mathbb{Q}_p)$ for all p.

This suggests that the fine-structure constant $\alpha \approx 1/137$ — a cross-ratio over $\mathbb{C}$ (the Archimedean place $p = \infty$) — should have counterparts $\alpha_p \in \mathbb{Q}_p$ for each finite prime p, with the product over all α_p forming the **adelic fine-structure constant.**

---

8. Summary of Results

8.1 Established

Result	Status
$\alpha = r_e/\bar{\lambda}_C$ is an algebraic identity	Verified
Cross-ratio is $\mathrm{PGL}(2)$-invariant over any field	Mathematical fact
Archimedean/non-Archimedean distinction is a "gauge choice" for invariants	Framework established
$?(x)$ is a groupoid isomorphism $\mathrm{PGL}(2, \mathbb{Z}) \to F$	Computationally confirmed
Tree projection produces heavy-tailed Euclidean distances	Computationally confirmed
Farey neighbor distances form a power law with infinite kurtosis	Computationally confirmed ($d \leq 2000$)
More tree structure $\to$ higher kurtosis	Computationally confirmed across 4 constraint levels
Primarity constraint reduces kurtosis from $\infty \to\sim 5$	Computationally confirmed
Bruhat-Tits building is the unified geometric object	Framework established

8.2 Open

Question	Status
Does a p-adic analog of α exist on the Bruhat-Tits boundary?	Theoretical conjecture
What is the adelic product formula relating α_p across primes?	Open
Can p-adic quantum mechanics be constructed on the building?	Research program
Does α have a discrete topological origin (linking number)?	Conjecture
Is the power-law exponent $\alpha \approx 1.484$ a universal constant?	Computational evidence; proof needed

---

9. Conclusion

We have presented a unified framework in which:

1. **The fine-structure constant** $\alpha = r_e/\bar{\lambda}_C$ is a process count — a projective invariant expressible as a degenerate cross-ratio.
2. **Cross-ratios** are the canonical $\mathrm{PGL}(2)$-invariants, defined identically over any field. The choice of Archimedean ($\mathbb{C}$) vs. non-Archimedean ($\mathbb{Q}_p$) field is a "gauge choice" — the invariant structure survives.
3. **The Minkowski** $?(x)$ is a groupoid isomorphism between $\mathrm{PGL}(2, \mathbb{Z})$ and the Thompson group F, with an explicit conjugation table. The cross-ratio is not pointwise preserved under this map, but transforms by the Thompson group cocycle.
4. **Tree-to-line projection** is a change of metric on the same underlying set. The Euclidean distance distribution of tree-adjacent points follows a power law with exponent $\alpha \approx 1.484$ and infinite kurtosis in the limit.
5. **The primarity constraint** on primes reduces this infinite kurtosis to approximately 4.9, producing a moderate-tailed distribution that matches the Cramér model. Prime gaps are the projection shadow of the divisibility tree through a primarity filter.
6. **The Bruhat-Tits building** — a $(p+1)$-regular tree for $\mathrm{PGL}(2, \mathbb{Q}_p)$ — is the unified geometric object underlying both Archimedean and non-Archimedean invariant structures. Its boundary supports cross-ratios, its chamber structure suggests a natural state space for non-Archimedean quantum mechanics, and the adelic product over all primes hints at the global invariant framework.

The cross-ratio is the canonical projective invariant — the simplest dimensionless quantity that survives changes of coordinate system and changes of field. Whether it is the **fundamental** invariant of structured reality, or one important invariant among many, is a hypothesis that can be tested, refined, and — as this investigation has shown — corrected when wrong. The arithmetic gauge principle, the projection mechanism, and the Bruhat-Tits geometry provide the framework for that investigation.

Mathematical objects: Cross-ratio, $\mathrm{PGL}(2)$, Minkowski $?(x)$, Thompson group F, Stern-Brocot tree, Bruhat-Tits building, Farey sequence, p-adic fields.

Physical objects: Fine-structure constant α, Compton wavelength, classical electron radius, Bohr radius, Cramér model for prime gaps.

Computational methods: Sieve of Eratosthenes, recursive Stern-Brocot splitting, exact rational arithmetic (Python `Fraction`), kurtosis analysis, log-log power-law regression.